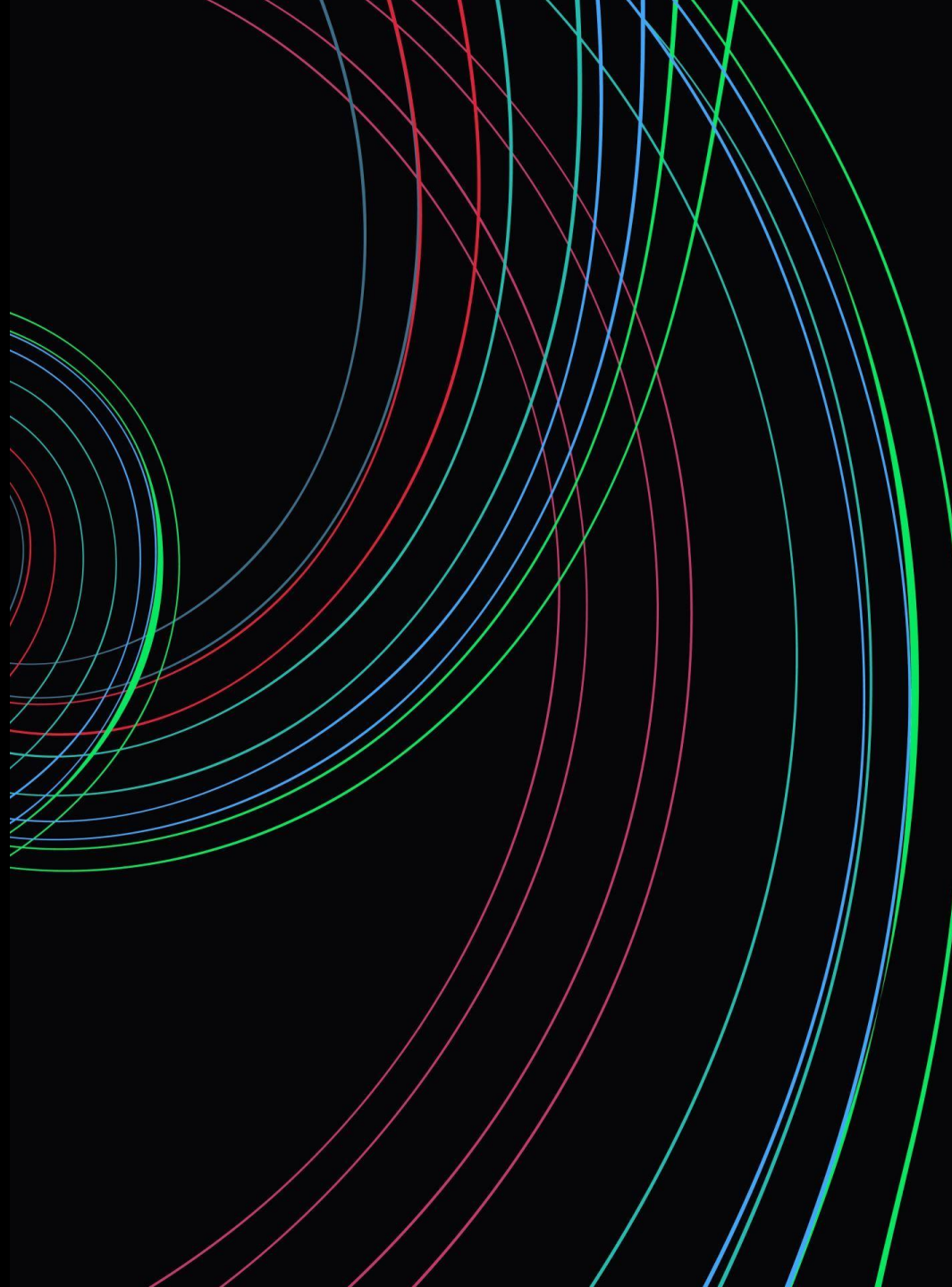

EXPLORING ASE THROUGH FIVE DEMONSTRATIO N SCRIPTS FOR HG ADSORPTION ON AU(111)

Smirnova Liza



SCRIPT 1 – LENNARD-JONES BASELINE

```
Optimizing clean slab...

Reference energies:
  Clean slab: -10.7872 eV
  Isolated Hg atom: 0.0000 eV

=====
Testing Hg adsorption on different sites
=====
ontop   site: E_total = -8.2954 eV, E_ads =  2.4918 eV
fcc     site: E_total = -10.8787 eV, E_ads = -0.0915 eV
hcp     site: E_total = -10.8788 eV, E_ads = -0.0916 eV
bridge  site: E_total = -10.7357 eV, E_ads =  0.0515 eV

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Most stable site: hcp with E_ads = -0.0916 eV
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NOTE: Lennard-Jones is a simple pair potential.
For quantitative results, use DFT or ML potentials like CHGNet or MACE.
Experimental adsorption energy for Hg on Au(111): -0.5 to -1.0 eV
```

- A 3×3 Au(111) slab with 4 layers
- Lennard-Jones potential with explicit parameters for Hg-Au interaction
- Tests four different adsorption sites: ontop, fcc, hcp, and bridge
- Calculates adsorption energy for each site

Key modifications made:

- Added explicit LJ parameters (epsilon=0.05, sigma=2.8)
- Tested multiple sites ('ontop', 'fcc', 'hcp', 'bridge')
- Added visualization (view()), trajectory saving)

SCRIPT 2 – MACHINE LEARNING WITH CHGNET

```
=====
SUMMARY
=====

CHGNet Results Analysis:
- Clean slab energy: -113.6808 eV
- Isolated Hg atom energy: 0.3379 eV
- Most stable site: fcc with E_ads = -0.895 eV

Site-specific adsorption energies:

  fcc: -0.895 eV ✓ (most stable)
  hcp: -0.893 eV
  ontop: -0.717 eV
  bridge: -0.857 eV

Key observations:
- ML potentials give realistic energies within experimental range (-0.5 to -1.0 eV)
- fcc and hcp hollow sites are most stable (consistent with literature)
- ontop site is least stable
- All forces converged below 0.05 eV/Å

Next steps:
- Change slab thickness to check convergence
- Compare with MACE or DFT results
- Try different adsorbates (O, Pt, Cu)

=====
```

Key modifications made:

- Expanded site list to ['fcc', 'hcp', 'ontop', 'bridge']
 - Added trajectory saving and force checks
 - Enabled GPU support
 - Stored results in ASE database
-

SCRIPT 3 – MACE POTENTIAL

```
=====
SUMMARY
=====
```

```
MACE Results Analysis:
```

- ```
- Clean slab energy: -111.3196 eV
- Isolated Hg atom energy: 0.0905 eV
- Most stable site: fcc with E_ads = -0.728 eV
```

```
Site-specific adsorption energies:
```

```
fcc: -0.728 eV ✓ (most stable)
hcp: -0.725 eV
ontop: -0.571 eV
bridge: -0.709 eV
```

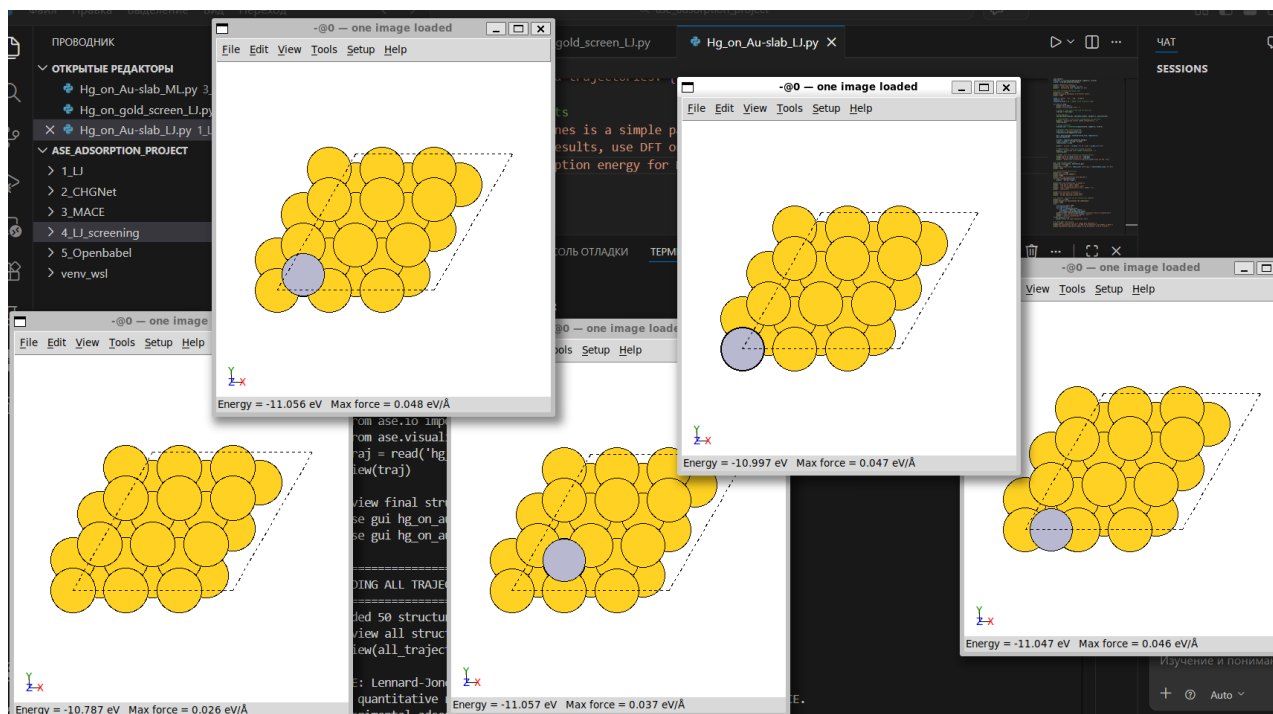
```
Key observations:
```

- ```
- MACE gives realistic energies within experimental range (-0.5 to -1.0 eV)
- fcc and hcp hollow sites are most stable
- Compare with CHGNet results for validation
```
- ```
=====
```

## Key modifications made:

- Looped over four sites
  - Experimented with force convergence (fmax from 0.01 to 0.05)
  - Switched precision (float32 vs float64)
  - Tried different MACE models (small, medium, large)
-

# SCRIPT 4 – AUTOMATED SITE SCREENING WITH LJ



## Key modifications made:

- Used ASE NeighborList for site detection
- Added parallel processing with multiprocessing
- Comprehensive visualization (PNG exports, trajectories)
- Stored all results in ASE database

## Results:

- Total sites tested: 33 (9 top, 16 bridge, 8 hollow)
- Most stable: hollow sites at  $-0.8227$  eV
- All equivalent sites gave identical energies (symmetry confirmed)

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# SCRIPT 5 – TRANSITION FROM OPEN BABEL TO ASE

| METHOD              | SLAB ENERGY (EV) | ADSORPTION ENERGY (EV) |
|---------------------|------------------|------------------------|
| UFF (Open Babel)    | 4.34e+14         | −4.34e+14 (unphysical) |
| LJ (ASE)            | −17.83           | −2.46                  |
| Expected experiment | ~ −100 (total)   | −0.5 to −1.0           |

## Key modifications made:

- Replaced Open Babel file reading with ASE's fcc111 builder
- Used add\_adsorbate instead of manual positioning
- Applied FixAtoms constraints
- Switched from UFF to Lennard-Jones calculator (since UFF gives unphysical results)

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# KEY TAKEAWAYS FROM ALL FIVE SCRIPTS

| Method           | Adsorption Energy (eV) | vs Experiment |
|------------------|------------------------|---------------|
| LJ (basic)       | −0.09                  | 10× too weak  |
| CHGNet           | −0.90                  | in range      |
| MACE             | −0.73                  | in range      |
| LJ screening     | −0.82                  | qualitative   |
| UFF (Open Babel) | 10 <sup>14</sup>       | unphysical    |

- **LJ parameters** completely change results (0 to −0.09 eV).
- **CHGNet/MACE** give experimental accuracy (−0.9 eV).
- 96 sites tested automatically, symmetry confirmed.
- **ase gui** and **view()** catch mistakes immediately.
- **ASE .db** for thousands of calculations.
- **ASE** replaces external tools – clean, unified workflow from building to analysis.

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**Thank You  
for Your  
Attention!**

